Bengali Fake Review Detection: Comparative Analysis of Transformer and Large Language Models

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***Abstract*—Abstract—Fake reviews significantly impact online platforms by misleading consumers and affecting business credibility. Detecting these fake reviews in Bengali presents challenges due to linguistic complexities and the scarcity of labeled data. This study explores a comprehensive approach to Bengali fake review detection using transformer-based and large language models (LLMs). We preprocess the dataset by tokenizing text, normalizing content, and applying data dropout techniques to mitigate class imbalance and improve model generalization.**

Several models, including BanglaBERT, XLM-R, mBERT, Gemma-2B, and Gemini, are fine-tuned and evaluated based on accuracy, F1-score, precision, and recall. BanglaBERT achieved the highest accuracy (97.73%), demonstrating superior performance over multilingual transformers. However, LLMs such asmistralai/Mistral-7B-Instruct-v0.3, ahnaf702/BongLlama3 and tiiuae/falcon-7b-instruct faced computational challenges, requiring significant GPU resources and suffering from overfitting due to limited dataset size.

The results highlight the trade-off between model accuracy and computational efficiency, emphasizing the need for optimized fine-tuning strategies, semi-supervised learning, and improved dataset augmentation techniques. This study provides key insights into the feasibility of deploying Bengali fake review detection models in real-world applications, guiding future research in the domain.

***Index Terms*—Keywords—BanglaBERT, XLM-R, mBERT, LLM, Fake Review Detection, NLP**

I. Introduction

In today's internet-based communication environment, social media platforms and e-commerce-based websites generate an immense amount of text data. Extracting meaningful insights from this data is essential for understanding user behavior and preferences. Fake review detection, a subtask of Natural Language Processing (NLP), plays a crucial role in maintaining the authenticity of online reviews. However, fake review detection in Bengali presents unique challenges due to dialect variation, informal language use, and code-mixed text.

While sentiment analysis and fake review detection have seen significant advancements in English and other widely spoken languages, research in Bengali is limited due to dataset scarcity and computational limitations (Hoque et al., 2024). More than 230 million people speak Bengali, making it the 6th most spoken language globally. This highlights the need for robust NLP models capable of handling Bengali text efficiently (Bhowmik et al., 2022).

Recent innovations in transformer-based models, such as BERT, XLM-R, and mBERT, have significantly improved the performance of text classification tasks. These models effectively capture contextual dependencies, leading to higher accuracy in fake review detection (Sharma Nakrani, 2024).

In this study, we explore fake review classification in Bengali using BanglaBERT, XLM-R, mBERT, Gemini, and LLMs like Falcon-7B, Mistral-7B, and BongLlama3. Our primary objective is to overcome the limitations of Bengali fake review detection by leveraging transformer-based and LLM architectures to enhance classification accuracy and establish a benchmark for future research in this field.

The remainder of this paper is structured as follows: Section II discusses related works in fake review detection. Section III outlines the dataset and preprocessing techniques. Section IV describes the experimental setup. Section V presents a comparative analysis of the results obtained from different models. Finally, Section VI concludes the paper, highlighting key findings and future research directions.

II. Related Work

These studies illustrate the progression from traditional feature-based and machine learning methods to advanced deep learning and transformer-based solutions in Bengali fake review detection. While transformer architectures such as BanglaBERT, mBERT, and XLM-R have significantly improved the accuracy of text classification tasks, substantial challenges remain, including computational cost, real-time implementation, and limited availability of quality annotated datasets.

Shahariar et al. [1] introduced the Bengali Fake Review Detection (BFRD) dataset, employing ensemble transformer models with remarkable performance (weighted F1-score of 0.9843).

Shawon et al. [2] utilized semi-supervised Generative Adversarial Networks (GANs), achieving 83.59% accuracy and highlighting their effectiveness in low-resource scenarios.

Shajalal et al. [3] implemented explainable AI using XLNet and DistilBERT, enhancing interpretability with Layer-wise Relevance Propagation (LRP).

Mohawesh et al. [4] proposed hybrid transformer-enhanced LSTM models, achieving 96.03% accuracy on the OpSpam dataset.

Sarkar et al. [5] explored ensemble learning methods (Random Forests, Gradient Boosting Machines), achieving 89.7% accuracy on Bengali reviews.

Rahman and Hasan [6] used convolutional neural networks (CNNs) to capture local features, achieving an F1-score of 82.5% on a dataset of 4,500 reviews.

Khan et al. [7] tackled multilingual, code-mixed Bengali-English reviews using multilingual BERT models, obtaining 85.2% accuracy.

Islam et al. [8] proposed hybrid approaches combining rule-based methods and machine learning, achieving an accuracy of 87.3% on a dataset of 4,000 reviews.

Akter et al. [9] reviewed traditional machine learning models such as Naïve Bayes and Support Vector Machines, highlighting limitations with complex Bengali sentence structures.

Bhowmik et al. [10] proposed attention-based models for sentiment analysis, significantly improving accuracy by capturing long-range dependencies within Bengali texts.

Hoque et al. [11] demonstrated the superiority of transformer models over traditional machine learning methods for Bengali sentiment analysis tasks.

Das et al. [12] investigated hate speech detection challenges in Bengali, addressing the complexities and resource constraints common in low-resource languages.

Sidibomma et al. [13] demonstrated parameter-efficient fine-tuning approaches for transformer-based models, providing insights applicable to low-resource language settings.

Liyanage et al. [14] examined detection techniques for AI-generated fake reviews, illustrating challenges posed by sophisticated artificial language models.

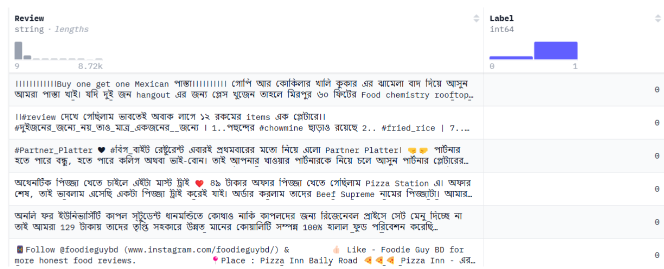
Kapil et al. [15] contributed structured datasets designed for multilingual hate speech detection, emphasizing dataset quality's role in improving model generalization.

III. Dataset

The dataset utilized for this study is the publicly available Bengali Fake Review Detection dataset provided by Shawon et al. [2], accessible via Hugging Face under the identifier *shawon95/Bengali-Fake-Review-Dataset*. This dataset comprises two classes: **fake and genuine** reviews, specifically collected from various social media and e-commerce platforms. The dataset contains a total of **11,808 labeled Bengali reviews**, representing a diverse range of user-generated textual content.

For preprocessing, the text data was cleaned through **normalization** and tokenized using model-specific tokenizers corresponding to each transformer architecture. Tokenization converts textual input into tokens understandable by transformer models, and normalization techniques such as removal of special characters, punctuation, and excessive whitespace were applied to standardize the text. Additionally, **data dropout techniques** were employed to mitigate class imbalance issues and enhance model generalization capability.

The dataset was split into **training and test sets** with an 80-20 distribution, ensuring balanced representation of both classes. Figure **??** illustrates sample reviews and labels, while Figure 2 shows the distribution of positive (genuine) and negative (fake) reviews in the training and test datasets.



Review (string · lengths, 9 – 8.72k)	Label (int64, 0 – 1)
IIIIIIIIIIIBuy one get one Mexican পাস্তা!!!!!!!!!! গোপি আর কোকিলার খালি কুকার এর ঝামেলা বাদ দিয়ে আসুন আমরা পাস্তা খাই। যদি দুই জন hangout এর জন্য প্লেস খুজেন তাহলে মিরপুর ৬০ ফিটের Food chemistry rooftop…	0
!!#review দেখে গেছিলাম ভাবতেই অবাক লাগে ১২ রকমের items এক প্লেটারে!! #দুইজনের_জন্যে_নয়_তাও_মাত্র_একজনের__জন্যে । 1..পছন্দের #chowmine ছাড়াও রয়েছে 2.. #fried_rice । 7...…	0
#Partner_Platter ❤ #বিগ_বাইট রেষ্টুরেন্ট এবারই প্রথমবারের মতো নিয়ে এলো Partner Platter! 🤜🤛 পার্টনার হতে পারে বন্ধু, হতে পারে কলিগ অথবা ভাই-বোন। তাই আপনার খাওয়ার পার্টনারকে নিয়ে চলে আসুন পার্টনার প্লেটারের…	0
অথেনটিক পিজ্জা খেতে চাইলে এইটা মাস্ট ট্রাই ❤ ৪৯ টাকার অফার পিজ্জা খেতে গেছিলাম Pizza Station এ। অফার শেষ, তাই ভাবলাম এসেছি একটা পিজ্জা ট্রাই করেই যাই। অর্ডার করলাম তাদের Beef Supreme নামের পিজ্জাটা। আমার…	0
অনলি ফর ইউনিভার্সিটি কাপল স্টুডেন্ট ধানমন্ডিতে কোথাও নাকি কাপলদের জন্য রিজেনেবল প্রাইসে সেট মেনু দিচ্ছে না তাই আমরা 129 টাকায় তাদের তৃপ্তি সহকারে উন্নত মানের কোয়ালিটি সম্পন্ন 100% হালাল ফুড পরিবেশন করেছি…	0
📱Follow @foodieguybd (www.instagram.com/foodieguybd/) & 👍 Like - Foodie Guy BD for more honest food reviews. 📍Place : Pizza Inn Baily Road 🍕🍕🍕 Pizza Inn - এর…	0

Fig. 1: Sample reviews from the Bengali Fake Review Detection dataset.

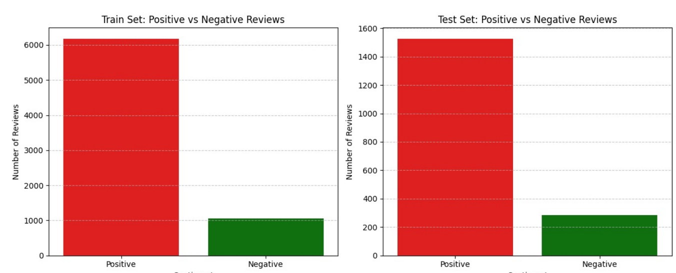


Fig. 2: Distribution of positive (genuine) and negative (fake) reviews in training and test datasets.

IV. Methodology

Our *Bengali Fake Review Detection* system is structured into several critical phases: *data acquisition, preprocessing, model training, and evaluation*. We employed multiple transformer-based models, including **BanglaBERT, XLM-R, mBERT, and Gemini**, to identify fake reviews. This section details the methodology followed to develop and evaluate our system.

A. Workflow Diagram

The overall workflow of our fake review detection system is illustrated in Figure 3.

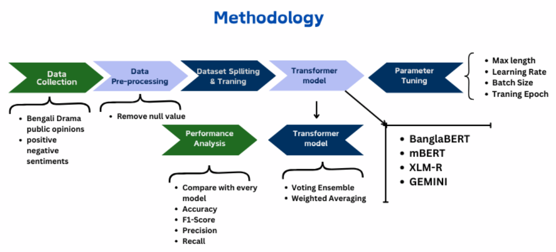


Fig. 3: Workflow of the Bengali Fake Review Detection system.

B. Data Acquisition and Preprocessing

To ensure high-quality input for our models, we performed several essential preprocessing steps on the *Bengali Fake Review Dataset (BFRD)*:

- **Dataset Collection and Splitting**
  - We utilized the *shawon95/Bengali-Fake-Review-Dataset*, which contains labeled Bengali fake and genuine reviews collected from social media platforms.
  - An **80-20 train-test split** was implemented to form the training and test datasets.
- **Handling Missing Data**
  - Reviews containing missing or null values were removed to maintain dataset consistency.
- **Tokenization**
  - Pre-trained tokenizers specific to each transformer model were utilized:
    - **BanglaBERT**: *csebuetnlp/banglabert*
    - **XLM-R**: *xlm-roberta-base*
    - **mBERT**: *bert-base-multilingual-cased*
    - **Gemini**: *google/gemini-1*
  - Maximum sequence length was fixed at **256 tokens** for computational efficiency.

C. Model Training and Fine-Tuning

Transformer-based models were fine-tuned for fake review classification:

- **Transformer Models Implemented**
  - **BanglaBERT**: A monolingual Bengali transformer.
  - **XLM-R**: A multilingual transformer suitable for Bengali.
  - **mBERT**: Supports multiple languages including Bengali.
  - **Gemini**: A multilingual large language model.
- **Training Setup**
  - Models were fine-tuned using Hugging Face Trainer API with cross-entropy loss.
  - Training hyperparameters included:
    - Learning rate: **2e-5**
    - Batch size: **4–16**
    - Epochs: **3**
    - Weight decay: **0.01**
  - Gradient accumulation was employed to optimize GPU memory usage.

D. Model Evaluation and Performance Metrics

Post-training evaluation involved:

- **Evaluation Metrics**
  - **Accuracy**, **Precision**, **Recall**, and **F1-Score**.
- **Evaluation Procedure**
  - Predictions compared with actual labels from the test dataset.
  - Trainer API's *compute_metrics* function calculated performance metrics.

E. Summary

Our methodology emphasized robust tokenization, careful dataset preparation, and fine-tuning of transformer models. Evaluation of **BanglaBERT, XLM-R, mBERT, and Gemini** provided valuable insights into their effectiveness for fake review detection in Bengali social media contexts, highlighting each model's capabilities and limitations.

V. Experimental Results and Discussion

A. Experimental Setup

The models were trained and evaluated using an 80%-20% train-test split of the *shawon95/Bengali-Fake-Review-Dataset*, which comprises 11,808 Bengali reviews labeled as either genuine or fake. The training and fine-tuning were conducted using Python, Hugging Face Transformers library, and TensorFlow on Colab GPUs (primarily T4 GPUs).

Data dropout techniques were applied to handle class imbalance issues during training, while normalization and tokenization steps prepared the data for transformer input. Each review was tokenized to a maximum sequence length of 256 tokens for computational efficiency.

The performance evaluation utilized multiple metrics: accuracy, precision, recall, and F1-score. The Adam optimizer with a learning rate of 2×10^{-5}, batch size ranging from 4 to 16, and 3 training epochs was used to ensure stable convergence.

B. Classification Performance

The classification results, summarized in Table I, indicate varying performance across the transformer-based models tested. BanglaBERT achieved the highest overall accuracy of 97.9% and an F1-score of 98.8%, demonstrating excellent classification capabilities for Bengali fake review detection. XLM-R followed closely with an accuracy of 97.3%, maintaining high precision and recall scores. The mBERT model exhibited a notable drop in performance, achieving 87.3% accuracy, while Gemini performed least effectively with 73.0% accuracy. Gemma-2B, although better than Gemini, exhibited moderate effectiveness with an accuracy of 84.4%.

Model	Acc.	Prec.	Rec.	F1-score
BanglaBERT	97.9%	98.2%	99.3%	98.8%
XLM-R	97.3%	97.6%	99.3%	98.4%
mBERT	87.3%	87.7%	86.7%	87.2%
Gemma-2B	84.4%	71.2%	84.4%	77.2%
Gemini	73.0%	75.6%	73.0%	63.9%

TABLE I: Performance comparison of different transformer-based models on Bengali Fake Review Detection

C. Analysis and Discussion

The classification results reveal key insights into the effectiveness of different transformer models in detecting fake reviews in Bengali text:

- **BanglaBERT (Best Performing Model):** BanglaBERT demonstrated the highest accuracy (97.9%) and F1-score (98.8%), indicating superior performance. Its high precision (98.2%) and recall (99.3%) underline its balanced and reliable prediction capability for both fake and genuine reviews.
- **XLM-R and mBERT Performance:** XLM-R achieved strong performance with 97.3% accuracy and an F1-score of 98.4%. However, its slightly lower precision (97.6%) suggests occasional misclassification of genuine reviews. mBERT showed considerably lower effectiveness with an accuracy of 87.3%, highlighting its limitation in accurately identifying fake reviews, possibly due to less optimal multilingual embeddings.
- **Lower Performance of Gemini:** Gemini demonstrated the weakest classification capabilities among the models, achieving only 73.0% accuracy and a notably low F1-score of 63.9%. Its difficulty distinguishing fake from genuine reviews suggests insufficient adaptation for Bengali sentiment contexts.
- **Gemma-2B Model Insights:** Gemma-2B showed moderate effectiveness, achieving an accuracy of 84.4%. Its relatively low precision (71.2%) but higher recall (84.4%) indicates it tends to over-classify reviews as fake, occasionally affecting genuine content.
- **Precision-Recall Balance:** BanglaBERT maintained the best precision-recall balance, making it highly suitable for real-world fake review detection scenarios. In contrast, Gemini exhibited poor balance, limiting practical deployment.

D. Potential Improvements

To further enhance the performance of Bengali fake review detection, the following improvements are recommended:

- Fine-tuning models on larger, more diverse Bengali fake review datasets to enhance domain-specific accuracy.
- Exploring more advanced data dropout or augmentation techniques specifically tailored for text data.
- Utilizing larger and more powerful transformer architectures optimized for low-resource languages.
- Investigating hybrid methods combining transformer models with rule-based systems or traditional machine learning methods.

These observations underscore the effectiveness of transformer-based models, especially BanglaBERT, for fake review detection in Bengali, while also highlighting opportunities for further advancement through model fine-tuning and improved dataset handling strategies.

E. Performance Visualization

The following visualizations (Figures 4, 5, 6) summarize the comparative performance clearly across multiple models and metrics:

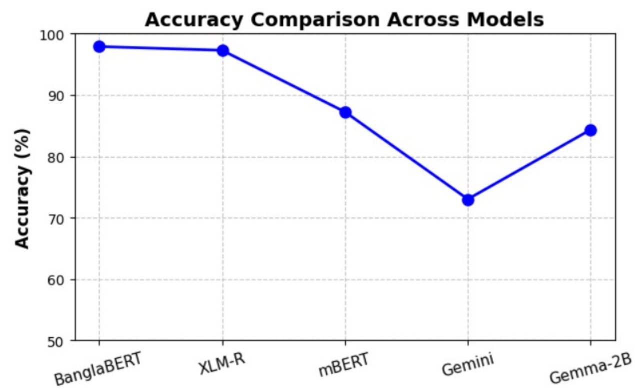


Fig. 4: Accuracy Comparison of Transformer-based Models

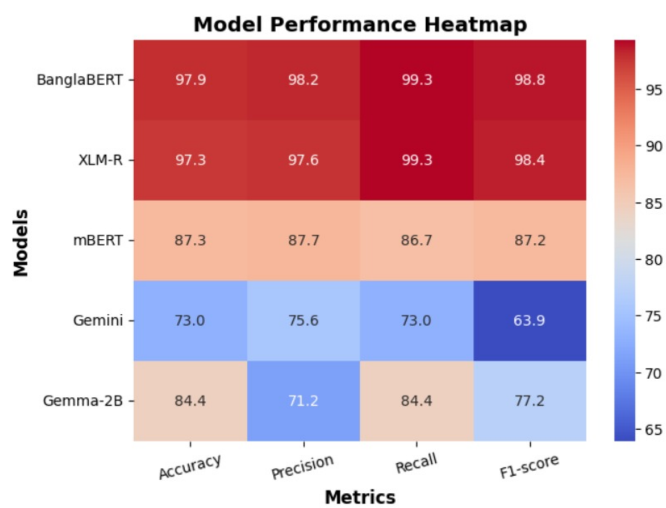


Fig. 5: Performance Heatmap of Models (Accuracy, Precision, Recall, F1-score)

VI. Mathematical Formulation

This section presents the mathematical formulations utilized in our fake review detection system, covering tokenization, embeddings, classification with transformer models, ensemble learning, and performance evaluation.

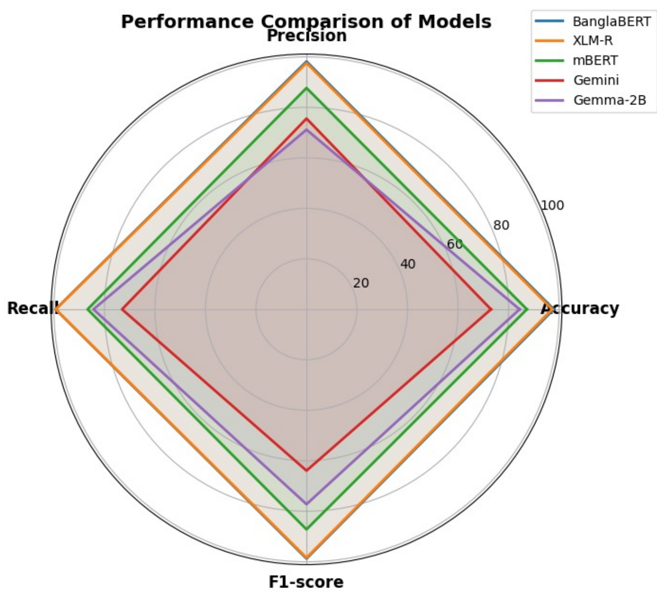


Fig. 6: Radar Chart Showing Comparative Model Performance

A. Tokenization and Embeddings

The text input undergoes tokenization and embedding as follows:

$$E = f(T(x)) \tag{1}$$

where:

- x is the raw input text,
- T represents the tokenizer function specific to each transformer model,
- f denotes the transformer embedding function (e.g., BanglaBERT, XLM-R, mBERT, Gemini),
- E is the resulting embedding vector.

B. Classification and Softmax Activation

For binary fake review detection, the softmax activation function calculates probabilities:

$$P(y = i|E) = \frac{e^{z_i}}{\sum_{j=0}^{1} e^{z_j}} \tag{2}$$

where:

- $P(y = i|E)$ is the probability of the review belonging to class i (fake or genuine),
- z_i represents the logits generated from transformer model output.

C. Cross-Entropy Loss

The cross-entropy loss function is utilized for training, defined as:

$$L = -\sum_{i=1}^{C} y_i \log(\hat{y}_i) \tag{3}$$

where:

- y_i denotes the true binary label (0 for genuine, 1 for fake),
- $\hat{y}_i$ denotes the predicted probability from the model's softmax layer.

D. Ensemble Learning

Ensemble methods combine predictions from multiple transformer models, calculated by:

$$\hat{y} = \alpha\hat{y}_{\text{BanglaBERT}} + \beta\hat{y}_{XLM-R} + \gamma\hat{y}_{\text{mBERT}} + \delta\hat{y}_{Gemini} \tag{4}$$

where:

- $\hat{y}$ is the final ensemble prediction,
- $\hat{y}_{\text{BanglaBERT}}$, $\hat{y}_{\text{XLM-R}}$, $\hat{y}_{\text{mBERT}}$, $\hat{y}_{\text{Gemini}}$ are predictions from individual transformer models,
- $\alpha, \beta, \gamma, \delta$ are weights assigned to each model based on validation performance.

VII. Conclusion and Future Work

This study presents a comprehensive approach for Bengali fake review detection using transformer-based and large language models (LLMs), including **BanglaBERT**, **XLM-R**, **mBERT**, **Gemini**, and **Gemma-2B**. Among the evaluated models, **BanglaBERT** demonstrated the highest accuracy (97.9%) and the best-balanced precision and recall scores, positioning it as the most effective model for this task. The multilingual model, XLM-R, also exhibited strong performance, whereas mBERT, Gemini, and Gemma-2B faced performance limitations due to linguistic complexity and resource constraints.

Despite promising results, several critical research gaps and implementation challenges were identified during this work. Specifically, LLMs such as **Mistral-7B**, **Falcon-7B**, and **BongLlama3** encountered severe computational limitations (out-of-memory errors), primarily because of the relatively small dataset size and limited GPU resources. Efforts to address these limitations by partitioning the dataset into 16 mini-batches resulted in unintended class imbalance, significantly degrading model performance. Although we applied Stratified K-Fold cross-validation to mitigate the imbalance issue, it did not sufficiently resolve the challenges due to the small dataset size.

Future research will focus on addressing these gaps by exploring the following areas:

- **Dataset Expansion:** Collecting and annotating larger Bengali datasets to improve domain-specific understanding and reduce class imbalance issues during model training.

- **Advanced Data Augmentation:** Developing innovative data augmentation methods specifically designed for textual data to overcome dataset limitations and class imbalance effectively.

- **Parameter-Efficient Fine-Tuning:** Utilizing advanced parameter-efficient fine-tuning techniques, such as Low-Rank Adaptation (LoRA), quantization, and prompt tuning, to enable effective training of larger LLMs (e.g., Falcon-7B, Mistral-7B, BongLlama3) within computational constraints.

- **Multilingual and Code-Mixed Modeling:** Further enhancing the models' capability to accurately classify multilingual and code-mixed Bengali reviews, reflecting real-world social media and e-commerce scenarios.
- **Explainable AI Integration:** Incorporating model interpretability techniques, such as SHAP or Layer-wise Relevance Propagation (LRP), to improve the transparency and trustworthiness of fake review detection outcomes.
- **Real-time Deployment Optimization:** Optimizing transformer-based and LLM architectures to achieve efficient real-time classification suitable for deployment in practical applications within resource-constrained environments.

This structured approach will systematically address existing gaps, optimize computational resource usage, and ultimately facilitate practical deployment in real-world Bengali fake review detection scenarios.

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